How is It Possible to Create a New Idea?

Stellan Ohlsson

University of Illinois at Chicago
Department of Psychology, Behavioral Sciences Building
1007 West Harrison Street, Chicago, IL 60607-7137
stellan@uic,edu

Abstract

Creativity requires the generation of novel ideas, a process often called insight. A model of this process is proposed. At the macro-level, the model is a standard symbolic architecture for problem solving. Its problem perception component consists of a layered network of processing units. At the micro-level, each such unit propagates its computational result along a subset of its forward links. The subset is selected by matching the activation levels of the links to a threshold for forward propagation. The activation levels are subject to feedback adjustment and re-distribution. A simple simulation demonstrates that these characteristics suffice to produce sudden alterations in a unit's output. The question whether a network consisting of such units will exhibit this behavior is raised.

Framing the Problem

Creativity stimulates interest in multiple disciplines (Boden 1994), but the fundamental question of how it is possible, in principle, to create something novel tends to be overlooked and there is no widely accepted answer. The human brain is a physiological system, governed by the causal laws of chemistry and physics. It is not clear how it can alter its own state and switch itself onto a different trajectory through its own state space. The conundrum applies with equal or greater force to computers.

The purpose of this paper is to put forward a theory of how a computational agent – brain, animal, machine or space alien, if any – can have a new idea. I assume without argument that the invention of something novel requires a new idea, using the term "idea" in its informal, vague sense, and that an explanation of how a new idea enters consciousness must be the keystone in any theory of creativity. Psychologists refer to this as the moment of *insight* (Ohlsson 1992). Such moments are reported in autobiographical accounts of creative processes (Hillier 1992), they can be observed in the laboratory (Metcalf 1986) and they can be captured in brain images Jung-Beeman et al. (2004).

The puzzling feature of insight is that the problem solving effort can undergo sudden alterations in mode and tempo: Steady exploration of the problem space (search) might degenerate into *perseverance* (continued pursuit of a solution already known to be unproductive) or end in an *impasse*, which in turn might be followed by insight (the sudden appearance of a relevant but previously dormant option) and continued search. If an agent is capable of solving a problem that requires a creative response, then what factor or factors prevent him, her or it from bringing that competence to bear and to make steady progress? Why impasses? If such factors are operating, how can the agent succeed eventually? Why do those factors not continue to operate, eventually resulting in terminal failure? The central fact to be explained is the possibility of eventual success in the face of initial failure.

The natural approach is to assume that there is one process operating during problem space search, often called "analytical problem solving" or "non-insight thinking", and a different process that is operating at the moment of insight, often called "restructuring." The latter can produce new ideas but the former cannot. This type of account requires an explanation why the creative process is not operating right from the start; if such a process is available, why does the agent not draw upon it immediately? Why do we not create continuously and effortlessly?

Also, a two-process account is incomplete without an explanation for how the creative, insight-producing process works. But this was the question at the start, so the breakdown into two types of processes (convergent vs. divergent; analytical vs. creative; crystallized vs. fluid; etc.) does not generate simpler or easier subproblems. But a good theory of insight should discharge all homunculi. That is, it should specify the act of creation in terms of a combination of processes that are not themselves creative. Paradoxically, the theory will be the stronger, the less exotic and the better understood those processes are.

The approach of the current work is to assume that search, perseverance, impasse and insight are different outcomes of one and the same set of computational processes. There is no special-purpose process that kicks in to break the agent out of an impasse or a case of perseverance. The differences in outcome are a function of how the basic processes interact under different conditions.

The model distinguishes two system levels, the macro (architecture) level and the micro (unit) level. The unit level specifies a particular type of processing component, to be called a *processing unit*, for lack of a better label. The set of operating rules for the units is the technical contribution of the present work. The architecture level

specifies the processing system within which the units are assumed to function. The overall architecture is not the focus of this paper, so I will merely list the features that I believe a computational system has to exhibit in order to have creative insights.

Macro-Level Architecture

The model assumes a symbolic architecture with a perceptual front end, a working memory, a long-term memory and an executive that carries out heuristic search by using the outputs from problem perception as cues with which to retrieve prior knowledge, which is then brought to bear on the current search state to select the best path forward. Within such an architecture, insight is an event in which the problem perception component alters its output; the agent "sees the problem in a new way" as we say. As a consequence, relevant but previously dormant knowledge is retrieved, effectively changing the problem space.

The key features that enable problem perception to alter its output, I propose, are representation, compositionality, layered processing, tentative propagation, feedback and forward re-propagation. All are familiar. What is new is the hypothesis that precisely these features characterize problem perception in creative agents. Brief explanations will suffice.

Representation. At the most fundamental level, it is possible to have a new idea because ideas are representations, a representation is always an interpretation of its referent and the referent does not uniquely determine its own interpretation. A revised problem representation is a re-interpretation of the reality confronting the agent, so a computational agent that is to have new ideas must operate on (local, symbolic) representations. Agents that are hooked into the environment via causal links rather than via representations do not have this capability. Moths fly into flames because their nervous systems implement a direct causal link between the intensity of the light hitting their eyes and the flapping rate of their wings. They cannot generate the idea of flying away from the light. Symbolic representations provide a measure of stimulus independence.

Compositionality. As psycholinguists and analytic philosophers have emphasized, cognitive representations consists of parts, smaller representations, combined in accordance with rules. Restructuring operates, I propose, by re-combining (simple) representations into different (complex) representations. The combinatorial character of creativity has been noted by creativity researchers from Henri Poincare to Arthur Koestler, and needs no defense or explanation.

Layered Processing. Combinatorial composition of representations requires at least two levels of processing, but I propose that a powerful creative agent needs many more. Human language works through words, phrases (clauses), sentences and discourse; visual perception works through at least features, objects and scenes. These course grained divisions no doubt underestimate the number of processing layers. Figure 1 illustrates the concept of successive processing layers. Each node corresponds to a processing unit. The bottom row of nodes can be thought of as feature detectors, while the top row can be conceptualized as chunks in working memory. In each cycle, each unit performs a computation and attempts to pass its product along to its recipients in the next layer up.

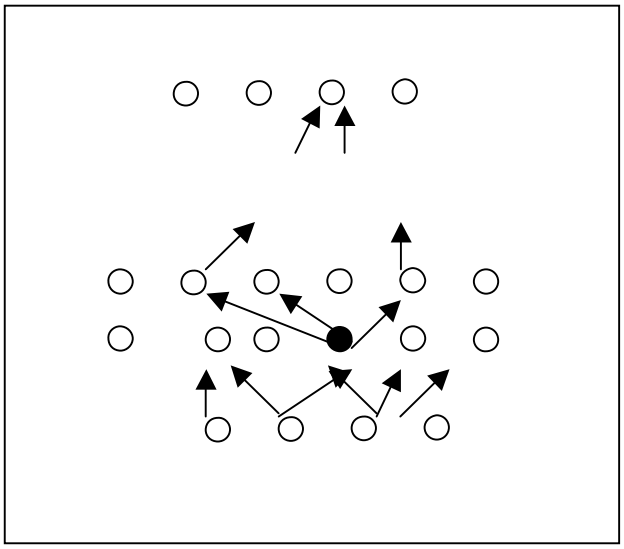

Figure 1. A partial view of a problem perception module.

Tentative Propagation. There is never sufficient information to decide with certainty the best response to a situation. A computational agent must be prepared to proceed even in the face of insufficient information, without guarantee that the action or conclusion produced is the best, or even good.

Feedback Links. Because products, conclusions and actions are necessarily tentative, their consequences must be evaluated. Outcome evaluations are feed back down to the lower layers of the problem perception component. The point of this is to alter the processing of the affected unit. S key question is how, exactly, the individual processing units are affected.

Forward Re-Propagation. When a processing unit is affected in such a way that it passes its computational product along a different set of forward links, the output of the problem perception component needs to be re-computed in order to take the change into account the change. Forward propagation along different links can be dampened or amplified as it progresses from layer to layer. In the former case, the change might fade without affecting the problem representation. In the latter case, it will cause

changes of greater and greater magnitude in each successive layer, eventually resulting in a revised problem representation.

The above properties describe the type of problem perception system within which the individual processing units are assumed to be operating. My hypothesis is that if we specify the individual processing units in a particular way, the possibility of suddenly change emerges.

Individual Processing Units

The task of a processing unit is to decide which of its outgoing links will forward the result of whatever symbolic computation the unit performs to the next layer. Fundamentally, a processing unit has to choose among the options for where to direct its output. The key difficulty is to specify its operation in such a way that one and the same unit can exhibit search, impasse, perseverance or insight, depending on the circumstances. In the context of the present model, search is indicated by the evaluation and pursuit of a set of options (links) available at the outset of the period of observation. Perseverance is the pursuit of an option (link) even though it generates negative feedback and a better alternative is available. Insight, finally, is the rise of a previously dormant option (link) above threshold.

General Description

A processing unit operates in a sequence of discreet *cycles*, N = 1, 2, 3,…, that define the *period of observation*. The operating cycles of all the processing units in the problem perception module are assumed to be synchronized. The observed cycles constitute a small segment of a longer sequence of cycles extending before and after the period of observation, as when a psychologist gives an experimental subject so many minutes to complete an insight problem.

Each processing unit is associated with an amount of *activation*, a transient quantity that fluctuates from cycles to cycle. A unit U receives activation in cycle N along its *input links* from those units in the immediately preceding processing layer that propagated their outputs to U during cycle N-1.

The unit activation in cycle N is distributed over the unit's *output links* in proportion to the activation levels they possessed during the previous cycle. If the activation received along each input link is constant throughout the period of observation, and if no other factor is operating, then there is no change in the unit activation or in the relative activation of the unit's output links.

An output link's share of the unit activation is *adjusted by feedback*. If the ultimate outcome of propagation along a link is desirable, positive feedback will be passed down the links that contributed to that outcome. The arrival of positive feedback increases the responsible link's share of the unit activation. If the outcome of propagation along the link is undesirable, negative feedback will be passed down. The arrival of negative feedback decreases a link's share of the unit activation.

The activation lost by one output link due to negative feedback is *re-distributed* across the other output links in proportion to their relative activation levels during the previous cycle. The consequence is that those links receive a greater share of the unit activation. Feedback might arrive along more than one link in any one cycle, so a link might receive activation from another unsuccessful but above-threshold link even as it gives up some of its own. Conceptually, the multiple, mutual feedback adjustments occur simultaneously.

The activation associated with an outgoing link after feedback adjustment is compared to a *threshold* for forward propagation. If the link activation is greater than the threshold, the computational product of the unit is propagated along that link. This will normally generate feedback. If the activation of the output link falls below the propagation threshold, that link does not propagate to the next layer. Its effective activation level is zero. Because it does not propagate any product, it is not instrumental in achieving the next outcome of the overall system, and hence does not generate feedback.

If a link starts out with above-threshold activation levels for some links but not for others, and the outcome of the overall system is positive, there will be no alteration in which subset of output links get to propagate, i.e., there will be no change in the problem representation. However, if the above-threshold links generate negative feedback, they will loose activation. Eventually, one or more of the previously dormant (below threshold) links might accrue enough activation through re-distribution to rise above threshold. Propagation along this previously unexplored link might affect the processing at higher layers.

Quantitative Model of a Single Unit

The purpose of the quantitative model is to demonstrate that feedback, redistribution and threshold suffice to generate the behavioral patterns of interest. Figure 2 shows a particular processing unit, corresponding, for example, to the filled-in node in Figure 1. It has two input links, In_1 and In_2, from its 'suppliers' in the immediately preceding layer. It has three output links, Out_1, Out_2 and Out_3, to 'recipients' in the next processing layer up. The model is abstract in that neither the nature of the symbolic inputs, the computation carried out on them nor the product of that computation are modeled, only the activation levels associated with each. The model is implemented in Excel.

The amount of activation in unit U in cycle N, a(U, N), is some function of the amount of activation arriving along its input links from the next layer down in cycle N-1. Addition across the input links will serve:

Eq. 1: $a(U, N+1) = \sum a(In_i, N)$, $i = 1, 2, 3,\ldots,$

where a(U, N+1) is the activation of unit U in cycle N+1 and $a(In_i, N)$ is the activation of input link i in cycle N. Equation 1 ignores lingering traces of past inputs. This allows the model to dispense with a decay function. This is a simplification.

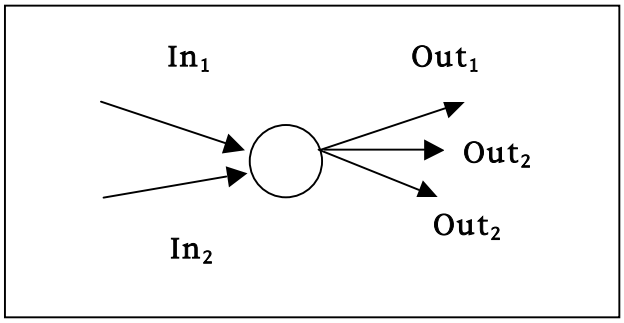

Figure 2. A single processing unit.

The unit activation is distributed across the outgoing links in proportion to their activation levels in the previous cycle, adjusted by feedback. The feedback generated by a given link is assumed to remain constant over the period of observation. I focus on the effect of negative feedback. The reduction of activation caused by negative feedback is assumed to be multiplicative, so feedback, f, is modeled by a number between 0 and 1, inclusive. The adjusted output activation adj(Out_i) of link i in cycle N is

Eq. 2: $\text{adj}(Out_i, N) = f \bullet a(Out_i, N-1)$.

If the feedback is 0, the link loses all its activation. If the feedback is 1, the activation is not adjusted, and if the feedback is larger than 1, the activation rises. The positive feedback case is not considered in this paper. In this single-unit model, there is no surrounding processing system, so the feedback is specified as one of the parameters given to the model (see Table 1). The value of the feedback varies from link to link, but for each link it is assumed constant throughout the period of observation, another simplification. In the runs reported, there is one output link, called the *correct* link and specified as having neutral feedback, $f = 1.0$, and two *incorrect* options, characterized by $f < 1.0$.

The adjusted activation determines an output link's share of the unit activation, *the effective activation*:

Eq. 3: $\text{eff}(Out_j, N) = \text{adj}(Out_j) / \text{Sum adj}(Out_i) * a(U, N)$, $i = 1,\ldots,j, \ldots$

The output link's share of the unit activation is compared to the threshold for forward propagation. The threshold is assumed to be constant during the observation period. Output links with above-threshold activation in cycle N cause feedback to be received along that link in cycle N+1.

In each operating cycle N, the unit receives its input from its suppliers, adds them up to determine the unit activation available in N (Eq. 1), distributes the unit activation across its output links, adjusts the activation for each output link by taking feedback into account, (Eq. 2), computes the effective feedback for that link (Eq. 3), selects the subset of output links that get to propagate in cycle N, and finally records that feedback is to be received along those links on the next cycle. The next operating cycle then initiates.

Quantitative Results

Table 1 shows the values of the input parameters for the 4 runs discussed in this paper. Figures 3-6 plot the effective activations of the output links in each cycle during a 25-cycle period of observation. Below-threshold links are shown as having zero activation. The threshold was set to 0.80. It is superimposed on each graph as a dotted line.

Table 1. Parameter values for four simulation runs.

Parameter	Run 1	Run 2	Run 3	Run 4
Threshold	0.80	0.80	0.80	0.80
In_1	2.00	2.00	2.00	2.00
In_2	1.00	1.00	1.00	1.00
Out_1	1.00	0.10	0.20	0.30
Out_2	1.00	1.70	1.20	1.00
Out_3	1.00	1.30	1.60	1.70
$f(Out_1)$	1.00	1.00	1.00	1.00
$f(Out_2)$	0.97	0.96	0.96	0.85
$f(Out_1)$	0.96	0.98	0.89	0.80

Figure 3 shows the result of run 1. In this case, all three output links were initially above threshold, and all three links remained above threshold for 11 cycles, i.e., all options available at the outset were pursued. At cycle 11, link 3 drops below threshold, and after another 10 cycles, link 3 drops out as well, leaving only link 1 above threshold. The activation of the latter gradually increases. This case corresponds to a mode of steady progress. The options available at the outset are evaluated, and the best one identified and pursued.

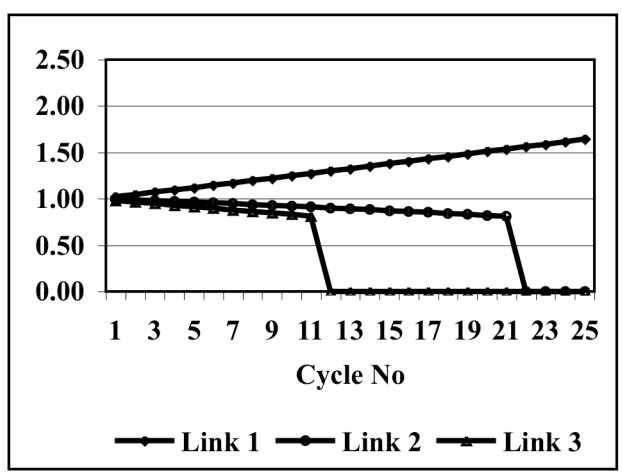

Figure 3. Output activation during run 1.

Figure 4 shows a different scenario. At the outset, the 'correct' (***f*** = 1.0) link 1 is below threshold, while the two 'incorrect' (***f*** < 1.0) options are both above threshold. As link 2 slowly looses activation and eventually drops below threshold, link 3, also 'incorrect', gradually accumulates activation, *even though it receives negative feedback on every cycle*. The activation lost by link 2 is redistributed over the other links, compensating for the loss of activation due to negative feedback for link 3. The dormant option also receives activation as a result of re-distribution, but not enough to rise above threshold during the period of observation. The result is that the processing unit keeps pursuing an inappropriate option even though there is a better alternative available. This is the type of behavior that we label perseverance.

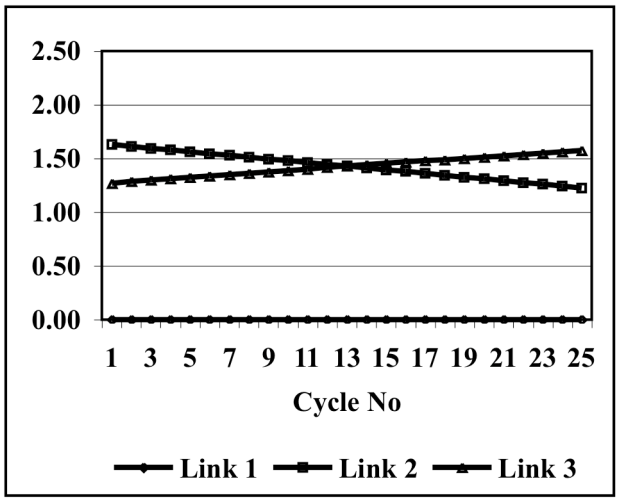

Figure 4. Output activation in run 2.

Figure 5 shows the insight case. At the outset, the correct option is once again below threshold, while the other two are active. As in run 2, one of the inappropriate links looses activation while the other becomes stronger. Eventually, in cycle 13, link 3 falls below threshold. In the period from cycle 13 to cycle 23, it appears that nothing is happening. The leading but incorrect option, link 2, continues to accumulate activation but both the other links are dormant. At the end of this period, the correct link rises above threshold, suddenly making a previously unheeded option available for consideration, the key characteristic of insight.

Figure 6 shows an alternative insight scenario. In this case, the rise of the dormant option happens earlier in the period of observation. Soon thereafter, both incorrect options fall below threshold. Once active, the correct option grabs all the unit activation.

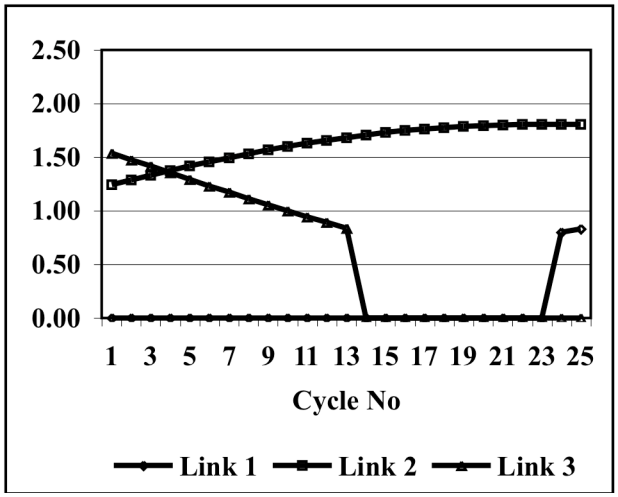

Figure 5. Output activation in run 3.

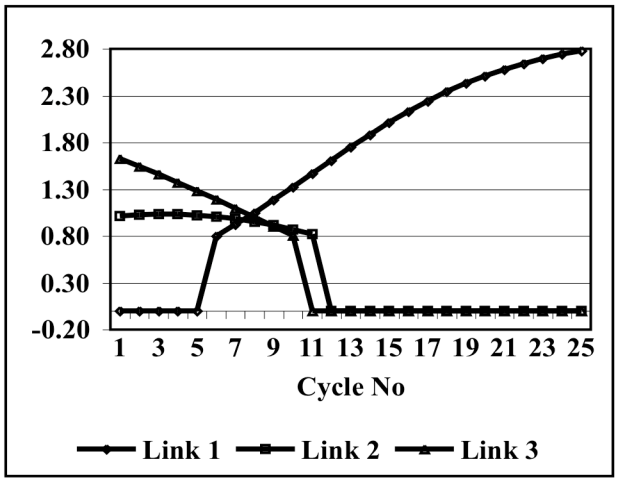

Figure 6. Output activation in run 4.

Discussion

The simulation runs illustrate that a single processing unit of the specified kind can exhibit qualitative alterations in mode and tempo as a consequence of a single set of processes operating continuously. Feedback-driven re-distribution of link activation in conjunction with a threshold produce search, perseverance and sudden change in the options considered (insight). The different outcomes are not associated with different processes, but are contingent on the values of the model parameters.

Copyright © 2007, Association for the Advancement of Artificial Intelligence (www.aaai.org). All rights reserved.

The model is simplistic in the extreme, consisting of 10 spreadsheet cells connected by Equations 1-3. Its advantage is that the relation between the operating principles and the resulting outcomes is transparent to intuition. Planned extensions include making feedback for above-threshold links a function of time (cycle number) so that the cost of pursuing an incorrect option increases over time, while negative feedback for below-threshold links dissipates. The activation of input links ought to fluctuate to reflect changes in the environment or in attention allocation. Both activation and threshold values ought to include random noise. Intuitive understanding will then have to yield to systematic parameter variations. How do the parameter settings affect variables like the number of cycles to insight?

In future work, multiple processing units of this sort will be connected in a layered network, to simulate the entire problem perception module (Figure 1). The key question is whether a network built out these kinds of processing units will exhibit the behaviors of interest. Do the distinctions between search, impasse, perseverance and insight scale up from a single unit to a network of hundreds or thousands of units? The intuition is that the forward propagation of a change in a single unit can spread through the system and cause a sudden change in its overall output, but only experiments with a model can illuminate the specific conditions under which this will happen.

A problem perception module must, in turn, be incorporated into a problem solving architecture. A change in problem representation does not necessarily imply a successful, creative solution to the problem at hand. The consequences of a new representation depend on the circumstances. If the new representation triggers the retrieval of relevant but previously dormant prior knowledge, then the problem solver is effectively working in a new problem space. However, this is not the only possible outcome. The new representation might trigger previously unheeded prior knowledge that is no more useful than the already active knowledge. Under these circumstances, there is an insight, but no success, at least not until some other insight occurs. Hence, the claim is not that the specified processes will guarantee successful solutions, but that they explain how a new option can come to mind after a period of failure.

Are the described mechanisms necessary for producing a creative response, or could the same effect be achieved by a qualitatively different system? Which component of the system outlined in this paper could we do without and still achieve a creative response? Unless the agent operates on interpretations, he or she cannot re-interpret a situation. Unless representations consist of parts, there is nothing to re-combine. But if representations are compositional, then there are multiple layers (at least two, part and whole). If a representation is to be changed in the face of failure, there must be some way to take feedback into account. If the change at a lower layer is to have any consequences, that change must be propagated upwards. It is difficult to imagine an agent that exhibits insight but lacks these macro features.

The uncertainty pertains to the exact design of the processing units. It is possible that such units can be designed on some other principles than re-distribution and threshold. Discussion has to await some alternative proposal.

In the meantime, the model described in this paper stands as an illustration of one relatively precise answer to why an agent might suddenly consider a previously dormant option after a period of unsatisfactory progress. Its main strength is that the different types of outcomes are not produced by qualitatively different processes, but by one and the same set of interacting processes running continuously. These processes are not in the least exotic, but well understood and pose no difficulties of implementation. All homunculi have been discharged.

References

Boden, M. A. ed. 1994. *Dimensions of Creativity*. Cambridge, MA: MIT Press.

Hillier, J. 1992. Electron Microscope and Microprobe Analysis: Recalling the Ambience of Some Inventions. In: R. J. Weber and D. N. Perkins, eds., *Inventive Minds: Creativity in Technology*. New York: Oxford University Press.

Jung-Beeman, M., Bowden, E. M., Haberman, J., Frymiare, J. L., Arambel-Liu, S., Greenblattt, R., Reber, P. J., & Kounios, J. 2004. Neural Activity When People Solve Verbal Problems With Insight. *PLoS Biology* 2: 0500-0510.

Metcalf, J. 1986. Feeling of Knowing in Memory and Problem Solving. *Journal of Experimental Psychology: Learning, Memory, and Cognition* 12: 288-294.

Ohlsson, S. 1992. Information-Processing Models of Insight and Related Phenomena. In: M. T. Keane & K. J. Gilhooly eds. *Advances in the Psychology of Thinking: Volume one*. New York: Harvester/Wheatsheaf.